



A Statistical Study of Housing Prices Using Descriptive Statistics, Correlation Analysis, and Multiple Linear Regression

Probability and Statistics

Section# F11

Submitted to:

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Introduction:

In this project, a statistical analysis is conducted on a housing dataset to study the relationship between various property features and house sale prices. The main objective of this analysis is to apply descriptive statistics, graphical techniques, correlation analysis, and regression modeling to understand how different variables influence housing prices.

The dataset used in this project is the **House Prices: Advanced Regression Techniques dataset**, which is publicly available on Kaggle. It contains detailed information about residential homes in Ames, Iowa, including physical characteristics, quality ratings, and sale prices. The dataset is widely used in data analysis and machine learning studies due to its richness and the presence of both numerical and categorical features.

For this project, a subset of **10 numerical variables** was selected for analysis. These variables were chosen based on their relevance to house pricing and statistical suitability. The dependent variable in this study is **SalePrice**, which represents the final selling price of a house. The independent variables used in the analysis are as follows:

- **SalePrice** (Dependent Variable): The final sale price of the house.
- **GrLivArea**: Above-ground living area in square feet.
- **OverallQual**: Overall material and finish quality of the house (rated from 1 to 10).
- **YearBuilt**: Original construction year of the house.
- **GarageCars**: Size of garage in terms of number of cars it can hold.
- **TotalBsmtSF**: Total square feet of basement area.
- **LotArea**: Size of the land lot in square feet.
- **1stFlrSF**: First floor area in square feet.
- **FullBath**: Number of full bathrooms above ground level.
- **TotRmsAbvGrd**: Total number of rooms above ground (excluding bathrooms).

The dataset contains over **1400 observations (houses)** and more than 80 variables in total. However, only the above 10 numerical variables were selected for this analysis to ensure clarity, manageability, and focus on key predictors of housing prices. Missing values were handled using listwise deletion, ensuring that only complete observations were used in the analysis.

This study applies fundamental statistical techniques including measures of central tendency (mean, median, mode), measures of dispersion (variance and standard deviation), quartile analysis, and interquartile range (IQR) for outlier detection. In addition, graphical methods such as histograms, boxplots, scatter plots, bar charts, line charts, and correlation heatmaps are used to visually explore the data.

Methodology:

This study was conducted using the House Prices: Advanced Regression Techniques dataset obtained from Kaggle. The dataset contains residential property information, including house characteristics and sale prices. Ten numerical variables were selected for analysis, with “SalePrice” as the dependent variable and “GrLivArea, OverallQual, YearBuilt, GarageCars, TotalBsmtSF, LotArea, 1stFlrSF, FullBath, and TotRmsAbvGrd” as independent variables.

- **Data Preprocessing**

The dataset was imported into Python using the Pandas library. Missing values were handled through listwise deletion using the `'dropna()'` method to ensure that only complete observations were included in the analysis.

- **Descriptive Statistical Analysis**

Descriptive statistics were computed for each selected variable to summarize the data. The measures calculated included:

Mean, Median, Mode, Variance, Standard Deviation, Quartiles and Interquartile Range (IQR)

Outliers were identified using the IQR method, where observations falling below $Q1 - 1.5(IQR)$ or above $Q3 + 1.5(IQR)$ were considered potential outliers.

- **Graphical Analysis**

Several graphical techniques like matplotlib lib were used to visualize the data:

Histograms with normal distribution curves to examine data distribution and normality.

Boxplots to detect outliers and visualize data spread.

Scatter plots to analyze relationships between the dependent variable and independent variables.

Correlation heatmaps to visualize relationships among variables.

- **Correlation Analysis**

Pearson correlation analysis was performed to measure the strength and direction of relationships between variables. Correlation coefficients range from -1 to $+1$, where values close to $+1$ indicate a strong positive relationship, values close to -1 indicate a strong negative relationship, and values near 0 indicate little or no relationship.

- **Regression Analysis**

Multiple Linear Regression was applied to model the relationship between the dependent variable (SalePrice) and the selected independent variables. The regression model follows the equation:

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

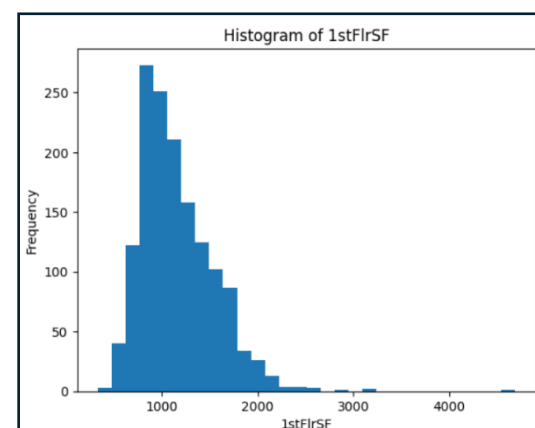
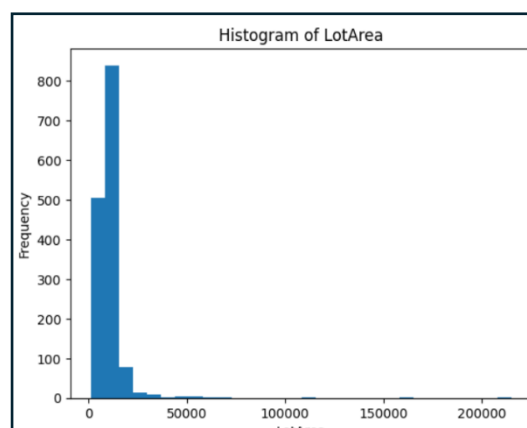
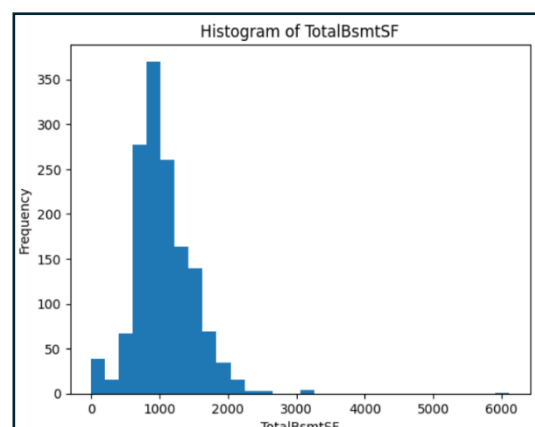
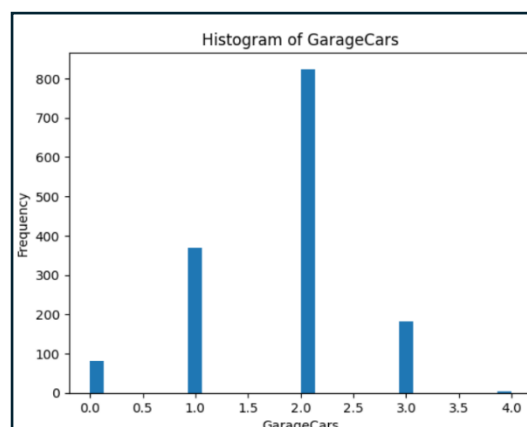
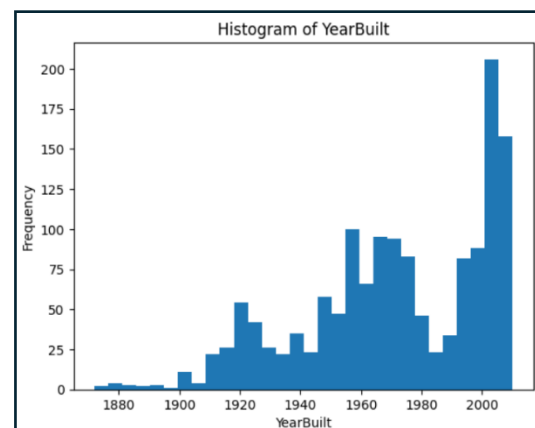
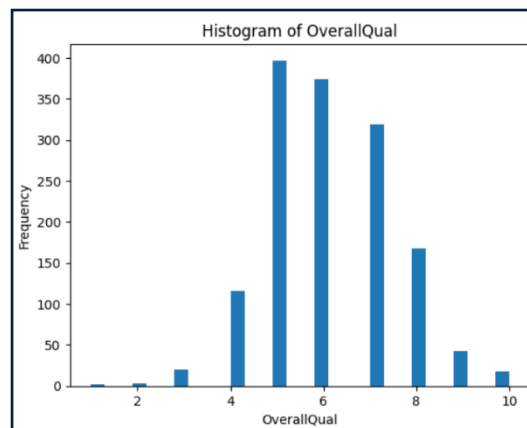
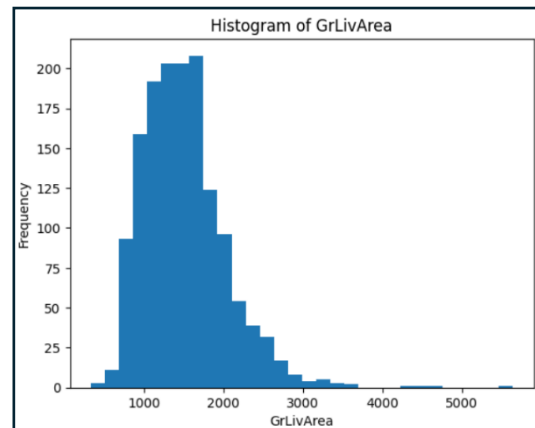
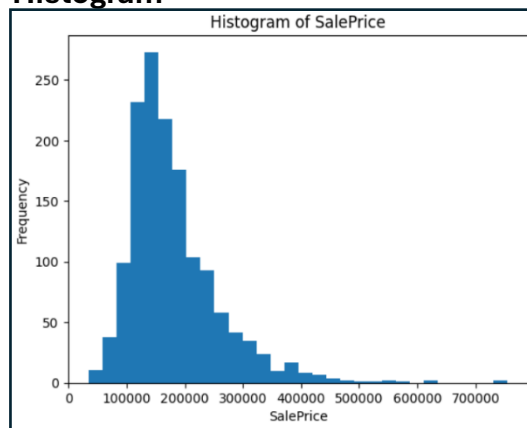
where Y represents SalePrice, α is the intercept, and β represents the regression coefficients of the independent variables. The model's performance was evaluated using the coefficient of determination (R^2), which measures the proportion of variation in house prices explained by the selected variables.

Descriptive Statistics:

Variables	SalePrice	GrLivArea	OverallQual	YearBuilt	GarageCars	TotalBsmtSF	LotArea	1stFlrSF	FullBath	TotRmsAbvGrd
Mean	180921.2	1515.5	6.1	1971.3	1.7671	1057.43	10516.83	1162.63	1.57	6.52
Median	163000	1464	6	1973	2	991.5	9478.5	1087	2	6
Mode	140000	864	5	2006	2	0	7200	864	2	6
Variance	6311111264.3	276129.633	1.913	912.215	0.5584	192462.4	99625649.7	149450.1	0.304	2.642
Std Dev	79442.5	525.5	1.4	30.203	0.7473	438.71	9981.3	386.6	0.551	1.6254
Q1	129975	1129	5	1954	1	795.8	7553.5	882	1	5
Q2	163000	1464	6	1973	2	991.5	9478.5	1087	2	6
Q3	214000	1776.75	7	2000	2	1298.3	11601.5	1391.3	2	7
IQR	84025	647.25	2	46	1	502.5	4048	509.3	1	2
Outliers	61	31	2	7	5	61	69	20	0	30

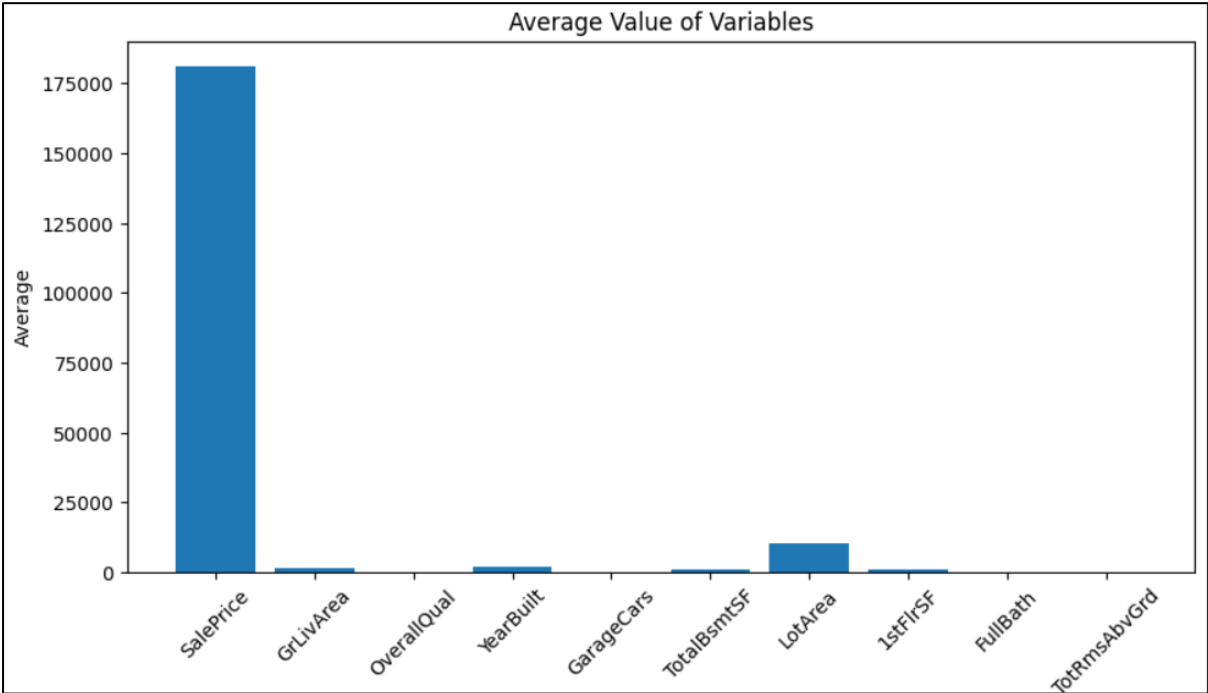
Graphical Representation:

Histogram

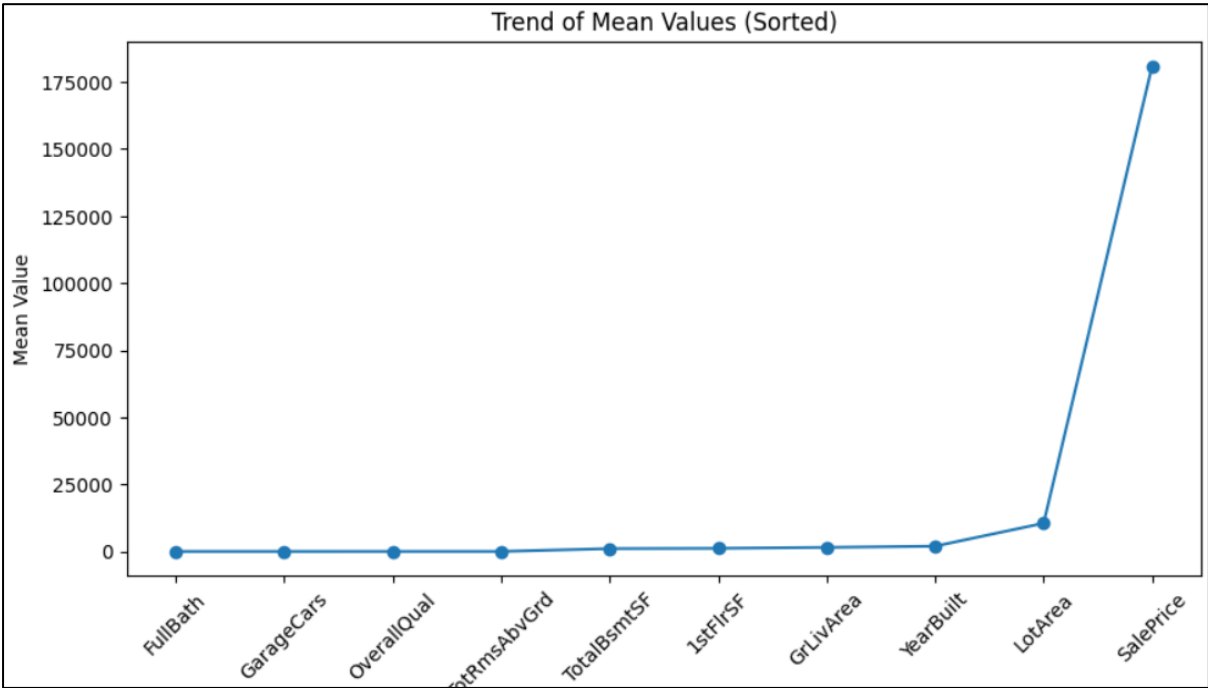


Histograms showed that variables such as SalePrice and LotArea are positively skewed, indicating presence of high-value outliers.

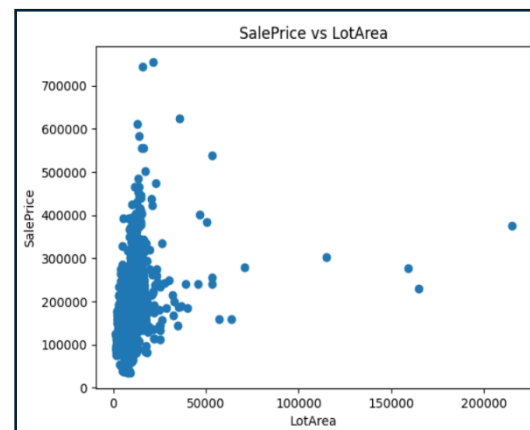
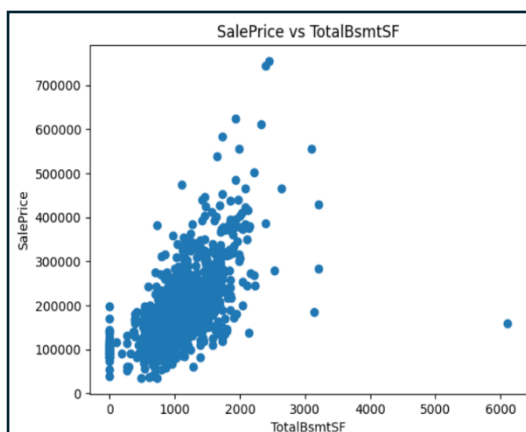
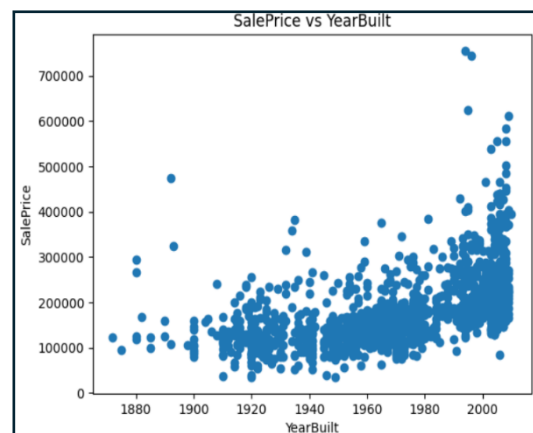
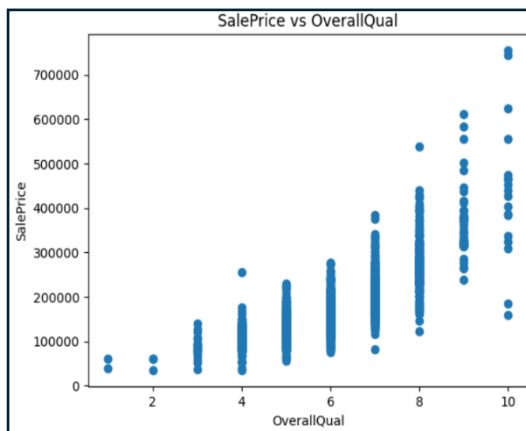
Bar Graph



Sorted Mean



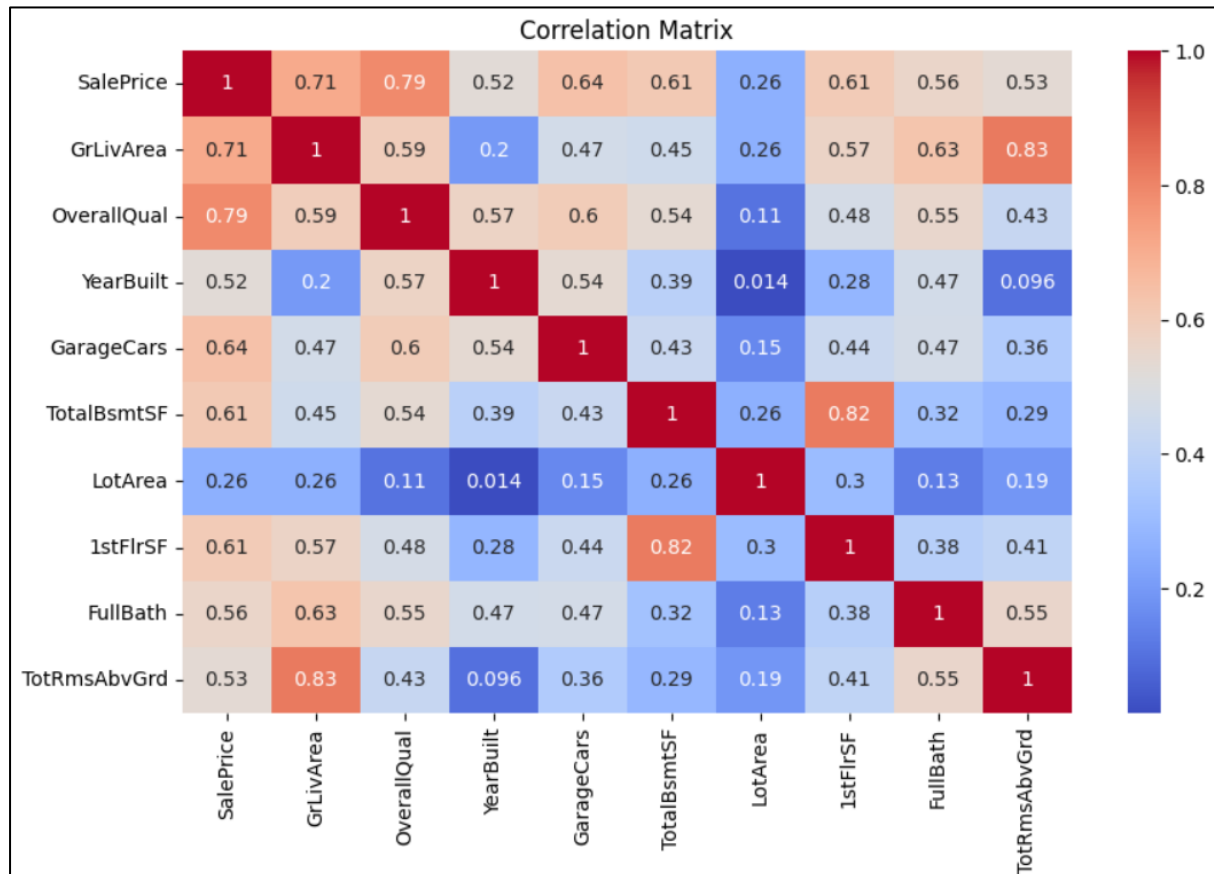
Scatter Plot:



Scatter plots showed that OverallQual and GrLivArea have a strong positive relationship with SalePrice, while LotArea shows a weaker correlation.

Corelation:

Heatmap:



+1 = strong positive relation

-1 = strong negative relation

0 = no relation

Correlation analysis shows relationships between variables ranging from -1 to +1. Strong positive correlation is observed between SalePrice and OverallQual and GrLivArea.

Regression:

Intercept (α): -789628.3060965422		
	Variable	Coefficient (β)
0	GrLivArea	48.997916
1	OverallQual	21594.308736
2	YearBuilt	360.031315
3	GarageCars	13434.319786
4	TotalBsmtSF	17.889110
5	LotArea	0.624858
6	1stFlrSF	12.175514
7	FullBath	-5596.583660
8	TotRmsAbvGrd	37.077684

The multiple linear regression model shows the relationship between house price and selected variables. OverallQual and GrLivArea have the strongest positive influence on SalePrice, indicating that house quality and living area are major price determinants. Some variables such as FullBath show a negative coefficient due to multicollinearity effects among features.

Most important features:

- OverallQual (strongest)
- GrLivArea
- GarageCars

Weak features:

- LotArea
- 1stFlrSF

Model insight:

- Not all features independently increase price
- Some overlap in information

R² Score Interpretation:

R² Score: 0.7753966836106277

The multiple regression model achieved an R² value of 0.775, indicating that approximately 77.5% of the variation in house prices can be explained by the selected independent variables. This suggests that the model has a strong predictive capability, while the remaining variation is due to other unobserved factors.

Conclusion:

This study analyzed the factors affecting house prices using descriptive statistics, graphical analysis, correlation, and regression techniques. The findings indicate that house quality (OverallQual), living area (GrLivArea), and garage capacity (GarageCars) are among the most influential factors affecting house prices. Correlation analysis confirmed strong positive relationships between these variables and SalePrice. The regression model achieved an R² value of 0.775, demonstrating good predictive performance. Overall, the analysis successfully identified key determinants of housing prices and provided valuable insights into the housing market.

Reference:

[Dataset link](#)